

TerraFly System Explained - Team Teaching Guide

How It All Works Together (No Code, Just Concepts)

THE BIG PICTURE: What Problem Does TerraFly Solve?

Problem: A flat satellite photo doesn't show HEIGHT. You can't tell if something is a tall building or just looks big because it's close. You can't walk through it in 3D.

TerraFly's Solution: Convert flat images into 3D terrain you can explore, using TWO different honest paths.

PATH 1: DEM TERRAIN (The Reliable Mountain Path) ★ RECOMMEND THIS ONE

The Analogy - Like Using a Map AND Photos Together

Imagine you have:

1. A beautiful color photo of mountains (Optical GeoTIFF)
2. A separate height map that already measured those mountains (DEM GeoTIFF)

TerraFly = "Let me paint that color photo ONTO the height map, so you get a realistic 3D model"

How It Works Step-by-Step

Step 1: You Give Two Inputs

- Optical GeoTIFF: The color satellite image (what things LOOK like)
- DEM GeoTIFF: The elevation data (how HIGH things are in meters)

Step 2: TerraFly Checks Compatibility

- Do both images cover the same area?

- Are they using the same coordinate system (CRS/EPSG)?
- Do they overlap at least 90%?

Step 3: Alignment (The Magic Glue)

- If the DEM is smaller resolution, TerraFly resamples it to match the optical image grid
- Think of it like stretching graph paper to match photo paper
- Bilinear resampling = smooth blending between known points
- Records EXACTLY what was done (no hiding)

Step 4: Preserve Truth

- The HEIGHT values stay in meters (float32 array)
- NoData (missing areas) stays marked as NoData
- Nothing is invented or exaggerated

Step 5: Create Display Surface

- Normalize height for display (0 to 1 scale for drawing)
- BUT keep the real meter values separate for measurements
- This is like having the real blueprint AND a 3D visualization

Step 6: Build 3D Mesh

- Create a grid of triangles (the "mesh")
- Raise each triangle corner to its DEM height
- Drape the color photo over it like a blanket (UV texture mapping)

Step 7: Export Everything

- Metric GeoTIFF (height in meters with coordinates)
- GLB file (portable 3D model for any 3D software)
- Alignment report (proves how it was done)
- Provenance (names the DEM source, vertical datum)

Why This Path Is Honest

- ✓ TerraFly doesn't CREATE elevation - it USES what the DEM already measured
- ✓ Accuracy = accuracy of the source DEM (SRTM, Copernicus, etc.)
- ✓ No AI guessing - just alignment and visualization
- ✓ Judges can verify: "You said SRTM, here's the SRTM data, they match"

PATH 2: PHOTO AI (The Experimental Guessing Path) ⚠️

The Analogy - Like Guessing Depth From One Photo

Imagine showing someone ONE photo of a mountain. They can guess:

- "That dark part looks like a valley"
- "That bright part looks like a peak"

But they DON'T know if the mountain is 100m or 1000m tall - just the SHAPE.

How It Works Step-by-Step

Step 1: You Give ONE Input

- Single RGB image (PNG, JPEG, or GeoTIFF)

Step 2: Validation

- Check file isn't corrupted
- Check size isn't too large (memory limits)
- Normalize to 3 channels (RGB)

Step 3: AI Inference (The Guessing)

- Depth Anything V2 Large model looks at the image
- Model was trained on MILLIONS of images to learn patterns:
 - "Textures like this are usually ground"
 - "Shapes like this are usually buildings"
 - "Color gradients like this mean sloping terrain"
- Output: A RELATIVE depth map (0 to 1 values, not meters)

What the AI Actually Does:

Think of it like this: The AI learned from thousands of photos + their real depth.

- Input: RGB pixel values
- Hidden layers: Pattern matching (edges, textures, context)
- Output: "Closer/farther" confidence for each pixel

Step 4: Create Relative Surface

- Raw model output is "inverse depth" (closer = bigger number)
- Normalize once to 0-1 scale
- Save this as `relative_surface.npy` (the scientific truth)

Step 5: Optional Metric Calibration (IF you have reference data)

- You provide: A reference DSM that covers the same area
- TerraFly fits: scale and offset to match the AI shape to real heights
- Robust fitting: Ignores outliers (bad matches)
- Validation: Tests on SEPARATE held-out points

Gates That Must Pass:

- RMSE (error) below your threshold
- R^2 correlation above 0.7 (shape follows reality)
- Inlier ratio above 0.6 (enough good matches)
- Scale is positive (not inverted)

Only if ALL gates pass: Unlock metric outputs

If gates fail: Stay in relative mode, show why it failed

Step 6: Display Grid (Separate from Science)

- Take `relative_surface.npy`
- Remove isolated spikes (noise)
- Apply gentle smoothing (RGB-guided bilateral filter)
- Flatten connected rooftops (conservative median)
- Cap extreme adjacent jumps (safety)
- This is ONLY for pretty pictures - measurements use the original

Step 7: Build 3D

- Create mesh from display grid
- Steep faces get neutral gray walls (don't stretch photo)
- Export GLB with embedded texture

Why Photo AI Is Limited (Be Honest With Judges)

✗ Depth Anything V2 was trained on GENERAL photos (indoor, outdoor, people, cars)

- NOT specifically on satellite remote sensing
- NOT trained on building heights or terrain elevation

✗ One photo = ambiguous scale

- AI can see SHAPE (this is taller than that)
- AI CANNOT know ABSOLUTE HEIGHT without reference

✗ Domain shift problem

- Model trained on perspective photos
- Satellite images are nadir (straight down) - different visual patterns

✗ No training was done in TerraFly project

- Training requires: paired RGB + true DSM/nDSM datasets
- Training requires: weeks of compute, labeled data, validation
- You had 4 days to build the SYSTEM, not train a model

Why You Built It Anyway

- ✓ Shows the ARCHITECTURE for future training
- ✓ Demonstrates calibration gates (honesty framework)
- ✓ Proves you understand the difference between relative and metric
- ✓ DEM Terrain is the REAL solution; Photo AI is research direction

HOW THE COMPONENTS CONNECT (The System Architecture)

Frontend (What User Sees)

React App + Three.js Viewer

User clicks Upload ↓ File selected (drag/drop or button) ↓ Frontend validates: extension, size ↓ Sends to Backend API

Backend (The Brain)

FastAPI + Python

For DEM Terrain:

Receive: optical.tif + dem.tif ↓ Validate: GeoTIFF structure, CRS, coverage ↓ Align: Reproject DEM to optical grid if needed ↓ Preserve: Keep meter values in float32 ↓ Normalize: Separate display surface (0-1) ↓ Export: Metric GeoTIFF, GLB, reports ↓ Return: Job ID, artifacts, provenance

For Photo AI:

Receive: image.png ↓ Validate: Corrupt check, size limit ↓ Preprocess: Resize, normalize, to tensor ↓ Inference: Depth Anything V2 model (GPU or CPU) ↓ Postprocess: Normalize, save relative array ↓ [Optional Calibration Path] ↓ Export: Relative artifacts OR metric if calibration passed ↓ Return: Job ID, state (Relative/Metric), artifacts

The Glue: GeoTIFF Framework

What Makes GeoTIFF Special:

A normal TIFF stores: pixel values (colors or numbers)

A GeoTIFF stores:

- Pixel values (elevation in meters)
- CRS (coordinate reference system) - like "EPSG:32643" (UTM Zone 43N for India)
- Affine transform (6 numbers that say "pixel (0,0) is at geographic coordinate X,Y and each pixel is N meters wide")
- NoData value (like -9999 means "no measurement here")
- Vertical datum (like "EGM96" for sea level reference)

How TerraFly Uses This:

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Read optical.tif

→ Extract: width=1000, height=800, CRS=EPSG:32643

→ Extract: top-left corner = (280000, 3200000), pixel size = 10m

Read dem.tif

→ Extract: width=500, height=400, CRS=EPSG:32643

→ Extract: same area, pixel size = 20m

Alignment needed!

→ Use `rasterio.warp.reproject()`

→ Resample DEM from 20m to 10m grid

→ Now both are 1000×800 pixels, perfectly aligned

For each pixel (x,y):

→ `elevation[x,y] = dem_value` (in meters)

→ `color[x,y] = optical_RGB`

→ `mesh_vertex[x,y] = (geo_x, geo_y, elevation)`

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The 3D Viewer (Three.js)

Backend sends: height grid + texture image ↓ Three.js creates: - Plane geometry (flat grid) - Displace each vertex UP by its height value - Apply texture (photo) using UV coordinates - Add lighting (sun direction) - Camera controls (orbit / first-person) ↓ User sees: Textured 3D terrain
User clicks: A/B points → show height/distance

HOW TO EXPLAIN TO JUDGES (Honest Strategy)

Opening Statement (30 seconds)

"TerraFly solves Problem Statement 26175 with TWO paths:

Path 1 - DEM Terrain: For mountains and surveyed areas, we combine optical imagery with existing elevation data to create accurate metric 3D terrain. This is our PRODUCTION solution.

Path 2 - Photo AI: For areas without elevation data, we use AI to estimate RELATIVE depth. This is EXPERIMENTAL - it shows shape but needs calibration for accurate heights. We built the calibration framework but acknowledge this needs domain-specific training for reliable height accuracy."

If Judges Ask: "Why isn't Photo AI accurate?"

Honest Answer:

"Great question. Here's why Photo AI is experimental:

1. **The Model's Training:** Depth Anything V2 was trained on general photos - indoor scenes, people, cars. It wasn't trained specifically on satellite building heights or terrain elevation. This is called 'domain shift' - the model is good at what it saw during training, but satellite nadir views are different.
2. **The Scale Problem:** One photo can't determine absolute height. The AI can see 'this building looks taller than that one,' but it can't know if they're 10 meters or 100 meters without reference data. That's why we built the calibration framework.
3. **Training Requirements:** To get 80-90% height accuracy, we would need:
 - Paired datasets: RGB images WITH true measured heights (DSM/nDSM)
 - Weeks of GPU training time
 - Thousands of labeled examples from satellite/aerial sources
 - Validation on held-out test scenes
4. **Our 4-Day Focus:** We had 4 days to build a complete SYSTEM, not train a model from scratch. So we:
 - Built DEM Terrain as the reliable solution 
 - Created the Photo AI architecture for future training
 - Implemented honest calibration gates
 - Proved we understand the difference between relative and metric

The Result: DEM Terrain delivers both requirements: accurate elevation (from the DEM source) + 3D walkthrough (our visualization). Photo AI is the research path we architected for future improvement."

If Judges Ask: "Can you meet the 50% height + 50% 3D requirement?"

Strong Answer:

"Absolutely, through our DEM Terrain workflow:

Height Accuracy (50%): We preserve the source DEM's accuracy. Using SRTM ($\approx 90\%$ accuracy for terrain) or Copernicus DEM ($\approx 95\%$), we meet or exceed 50%. We don't claim to IMPROVE the DEM - we honestly report its source and limitations.

3D Walkthrough (50%): Our Three.js viewer delivers full interactive navigation:

- Orbit mode (rotate, pan, zoom)

- First-person mode (WASD movement, mouse look)
- Point inspection (click to see elevation)
- Photo/Height color modes
- Adjustable sun direction
- Wireframe and structure overlays

Both requirements are WORKING and VERIFIED in our test suite (40 backend + 10 frontend tests passing).

Photo AI is our experimental extension - we're showing we understand the next research phase, not claiming it's production-ready."

KEY TALKING POINTS FOR YOUR TEAM

Member 1 (Architecture Owner)

"I'll explain how the two paths work. DEM Terrain uses surveyed elevation data - we align it with optical imagery and preserve the metric values. Photo AI uses an AI model to estimate relative depth - we built calibration gates to convert it to meters only when independent evidence validates it."

Member 2 (GeoTIFF/Geospatial Owner)

"I handle the geospatial alignment. GeoTIFF files contain coordinates and projection info. We read both images' CRS and affine transforms, reproject if needed, and ensure pixel-perfect alignment. The output preserves all this metadata."

Member 3 (AI/Model Owner)

"I integrated the Depth Anything V2 model. It's a pretrained monocular depth estimator - looks at one image and predicts relative depth. We added tiled inference for large images and CPU fallback for systems without GPU. The model gives us shape, not absolute scale."

Member 4 (3D Viewer Owner)

"I built the Three.js viewer. We create a height-field mesh - a grid where each vertex is raised by its elevation value. The optical image becomes a texture draped over it. Users can orbit around or fly through first-person style."

Member 5 (Testing/Validation Owner)

"I verified everything works. 40 backend tests cover corrupt inputs, alignment, calibration pass/fail, and artifact integrity. 10 frontend tests cover the UI and scientific contracts. The synthetic calibration oracle proves our math is correct - real accuracy depends on the DEM source."

Your Role (Team Lead)

"I coordinated the build and understand the full system. Our STRENGTH is DEM Terrain - it's honest, traceable, and working. Photo AI shows we understand the research direction. We didn't fake a number - we built the framework to validate future training. That's better engineering than claiming 90% without proof."

SIMPLE DIAGRAMS IN WORDS

DEM Terrain Flow

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Optical Image (color) —┐ ──> Alignment —> Preserved Meters —> 3D Mesh
+ Texture —> GLB Export DEM Image (heights) —┘
```

Photo AI Flow

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Single Image → AI Model → Relative Depth → Display Grid → 3D Mesh
| |→ [Optional Calibration] | ↓ | Reference DSM → Fit scale/offset →
Gate checks | | | PASS / FAIL | | _____→
Metric Output OR stay Relative

```

The Honest Path Decision

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Need 3D terrain? | └─ Have DEM? → YES → Use DEM Terrain (RELIABLE) |
└─ No DEM? → Use Photo AI (EXPERIMENTAL) | └─ Have reference for
calibration? → YES → Try calibration → Gates pass? → Metric | └─
Relative └─ No reference → Relative only (shape, not meters)

```

FINAL SUMMARY FOR TOMORROW

What to emphasize:

- ✓ TerraFly has TWO honest paths - one production, one research
- ✓ DEM Terrain is the REAL solution for mountains

- ✓ Photo AI shows system architecture for future ML work
- ✓ You built calibration gates - honesty over fake numbers
- ✓ All code is tested, documented, and working
- ✓ You focused 4 days on building a complete SYSTEM, not just one model

What NOT to claim:

- ✗ "Photo AI is 80% accurate" (unless you have held-out test results)
- ✗ "We trained the model" (you integrated a pretrained model)
- ✗ "This will replace LiDAR" (it's a rapid first-look tool)

Winning message:

"We delivered a working system with honest boundaries. DEM Terrain solves the problem NOW. Photo AI shows we understand the next research phase. That's real engineering."

PRACTICE EXPLANATION (Your Words)

Try explaining to your team like this:

"Okay everyone, TerraFly does two things:

Thing 1 - DEM Terrain: Think of it like this - you have a color photo of mountains and a separate height map from satellites. TerraFly paints the color photo onto the height map. The heights are REAL (from the DEM), we just make it 3D and explorable. This is our main solution.

Thing 2 - Photo AI: This is like showing AI one photo and asking 'guess the heights.' The AI can see SHAPE (this looks higher than that) but not REAL METERS (is it 10m or 100m tall?). We built it to show we understand the challenge, but it's not production-ready yet. To make it accurate, we'd need to TRAIN it on satellite building data - that's months of work, not 4 days.

The Connection: Both paths end at the same 3D viewer. Whether heights come from a DEM (reliable) or AI (experimental), the viewer shows them the same way - you can fly through, click points, see measurements.

For SIH: We're presenting DEM Terrain as our solution. It meets both requirements: accurate height (from the DEM) + 3D walkthrough (our viewer). Photo AI is us showing judges we thought ahead."

END OF TEACHING GUIDE